CASCADIA ANALYTICS

Synopsys

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M&A Integration Masking Organic Valuation Reality

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Synopsys commands an entrenched position as the dominant chip design automation software vendor, yet current market pricing reflects aggressive expectations that overlook significant post-merger integration costs and balance sheet leverage. While headline revenues have expanded rapidly following the Ansys acquisition, organic momentum in core segments has faced friction from interest burdens and substantial GAAP-to-non-GAAP differentials. We believe the current market valuation at $490.19 fails to account for intermediate cash flow realities and capital structure constraints. We initiate coverage with a Hold (Valuation Concerns) rating, establishing a probability-weighted blended target of $341.74, representing -30.3% upside.

HOLD	$341.74	-30.3%	$490.19
Hold — Valuation Concerns	BLENDED TARGET	BLENDED UPSIDE	REFERENCE PRICE
INVESTMENT RATING			

While equity markets celebrate top-line expansion and backlogs reaching $11B to $11.4B following the Ansys transaction, our forensic analysis highlights that high leverage of $14.3B in total debt and quarterly interest expense reaching $133M to $195M present persistent per-share earnings dilution. The market continues to price Synopsys as an unencumbered high-multiple software asset, whereas we identify substantial valuation risk stemming from real amortization, restructuring costs, and integration execution.

Executive summary

Synopsys operates as the dominant chip design automation (EDA) software vendor, selling licenses chipmakers need to design silicon. The core business spans Design Automation and Design IP, providing mission-critical infrastructure across modern computational architecture. In fiscal year 2025, Synopsys generated $7.05B in revenue, delivering a gross profit of $5.43B and an operating income of $914.9M, alongside net income of $1.33B and free cash flow of $1.35B. While gross margins remained structurally sound at 77.0%, GAAP operating margins settled at 13.0%, reflecting significant integration and amortization expenditures. Free cash flow margin reached 19.1%, supporting ongoing debt obligations, while return on equity finished at 4.7%.

Recent quarterly performance illustrates both the structural power of the core franchise and the operational friction introduced by large-scale M&A. In Q4 FY2025, headline revenue reached $2.26B, up 38% year-over-year, yet $668M came from newly acquired Ansys, implying organic revenue contracted ~3% to ~$1.59B as Design IP revenue declined 21% with margins dropping to 13.8%. In Q1 FY2026, revenue rose 65.6% year-over-year to $2.41B, but GAAP net income plummeted 78% to $65M under the weight of $404M in amortization and $118M in restructuring charges tied to a 10% workforce reduction. Q2 FY2026 demonstrated top-line acceleration to $2.276B, up 42% year-over-year, yet non-GAAP EPS contracted 9% to $3.35, pressured by $133M in quarterly interest expense and dilution from 36M new Ansys shares.

Operational execution inflected positively in Q3 FY2026, where revenue reached $2.48B and non-GAAP EPS hit $3.91, both exceeding guidance. Design IP staged a noticeable recovery, advancing 10.8% year-over-year to $473.8M, with segment adjusted operating margins expanding to 26.5% from 20.1% in the prior year. Management raised full-year FY2026 targets across revenue, margin, EPS, and free cash flow, projecting $2.98B in Ansys revenue. Nevertheless, substantial structural divergence persists between non-GAAP metrics and GAAP results, evidenced by a 27-point margin gap in Q3 FY2026 where GAAP operating margin stood at 14.4% compared to 41.6% non-GAAP, driven by $402M in amortization and $232M in stock-based compensation.

From a balance sheet perspective, the enterprise absorbed substantial leverage to execute its strategic roadmap. Fiscal 2025 closed with total debt of $14.3B against cash of $2.89B, generating net debt of $11.4B. Although operating cash flow rebounded sharply in Q1 FY2026 to $856.8M, compared to an outflow of $67.5M in the prior-year period, debt servicing remains a primary call on capital. Management authorized a $2.0B share repurchase program alongside debt paydown initiatives. However, with market capitalisation standing at $94.3B, trading at an EV to latest fiscal year EBITDA multiple of 42.3x and a P/E multiple of 61.0x on latest fiscal year diluted EPS of $8.04, current valuation levels leave no margin for execution missteps.

Operational snapshot

$490.19	$7.05B	$1.35B	19.1%	$11.4B	$490.19
REF. PRICE	FY2025 REVENUE	FY2025 FREE CASH FLOW	FCF MARGIN	NET DEBT	REF. CLOSE

Fiscal-year figures as reported in company filings and collected by Cascadia Analytics; ref. price as of 2026-10-29.

The moat is Pervasive and Irreplaceable

The enterprise foundation of Synopsys rests on profound switching costs embedded across global semiconductor design workflows. As the dominant chip design automation software vendor, the company sells licenses chipmakers need to design silicon, embedding its proprietary algorithms into standard design methodologies. Semiconductor engineers spend careers building proprietary custom scripts and design flows around Synopsys toolchains. Migrating an established advanced-node project to an alternative platform risks timing errors, unverified silicon tape-outs, and catastrophic mask redesign costs. Consequently, enterprise customer retention remains near total across primary semiconductor leaders, providing unparalleled operational stability.

Regulatory and technological barriers further reinforce this structural moat. Advanced semiconductor physics require co-optimization of logic, physics, thermal management, and power dissipation. The combination of Design Automation tools with Ansys multi-physics simulation assets broadens the software footprint across silicon-to-systems engineering. Design Automation sustained an elite 45.2% adjusted operating margin in Q3 FY2026, after achieving 47.3% in Q1 FY2026 and 41.5% in Q4 FY2025. This enduring profitability demonstrates deep pricing power, allowing the firm to capture value even as chip complexity compounds geometrically under modern artificial intelligence workloads.

In addition, the Design IP portfolio serves as a strategic moat enhancer. By licensing standardized subsystem blocks, memory interfaces, and interconnect intellectual property, Synopsys reduces time-to-market for merchant silicon vendors and internal design teams. Although the IP business experienced transition volatility, including a segment margin collapse to 13.8% in Q4 FY2025 and 16.2% in Q1 FY2026 during a shift toward chiplet subsystems, segment adjusted operating margins recovered to 26.5% in Q3 FY2026 on revenue of $473.8M. The technical integration between foundational EDA software licenses and proven IP blocks forms an ecosystem barrier that competitors struggle to duplicate.

Valuation

Three scenarios make the valuation range explicit

Our valuation framework synthesizes three discrete operational scenarios across the FY2026 to FY2031 forecast window, utilizing an explicit EBIT-bridge discounted cash flow model. We apply a weighted average cost of capital of 9.14% and a base-case terminal growth rate of 3.0%. Diluted shares in the model are established at 192.3 million. The resulting probability-weighted distribution reveals that the reference market price of $490.19 significantly overstates long-term fundamental realization across the probability distribution.

The model establishes our Base scenario at $335.92 per share, reflecting a 50% probability weighting. This path models a 10.7% revenue compound annual growth rate through FY2031 and an expansion in target free cash flow margin to 28.0% by the terminal forecast year, stepping up from the FY2025 level of 19.1%. The Bull scenario, assigned a 25% probability, yields a target of $466.64 based on a 13.2% revenue CAGR and a 32.0% terminal FCF margin. The Bear scenario, weighted at 25%, generates a valuation of $228.48, assuming a 7.9% revenue CAGR and a 24.0% terminal FCF margin.

Probability weighting these outcomes generates a blended target of $341.74, indicating -30.3% downside from the current market price of $490.19. The asymmetric downside skew highlights that even under our optimistic scenario, the intrinsic equity value of $466.64 resides 4.8% below the current trading level, illustrating the degree to which current multiples incorporate flawless forward execution.

SCENARIO	WEIGHT	REVENUE CAGR	TARGET FCF MARGIN	TARGET	UPSIDE
Bear	25%	7.9%	24.0%	$228.48	-53.4%
Base	50%	10.7%	28.0%	$335.92	-31.5%
Bull	25%	13.2%	32.0%	$466.64	-4.8%

Bear $228.48

This scenario models prolonged integration friction where ongoing interest costs near $133M to $195M per quarter consume cash flows while divestitures including Optical, PowerArtist, and Processor IP create a lingering drag. Design IP

Base $335.92

Our base case reflects a 10.7% revenue CAGR as Ansys delivers on its expected $2.98B FY2026 contribution and Design Automation maintains margins above 45%. Free cash flow margins normalize toward 28.0% as debt paydown mitigates

Bull $466.64

Under this pathway, silicon-to-systems demand accelerates EDA revenue expansion to a 13.2% CAGR while aggressive synergy realization pushes terminal FCF margins to 32.0%. Rapid deleveraging and robust execution across the $11B to $11.4B

margins fail to fully recover from prior troughs, dampening top-line CAGR to 7.9% and capping target FCF margins at 24.0%.

annual interest expense, yielding an intrinsic value of $335.92.

backlog drive equity valuation to $466.64.

Valuation framework

A live discount rate, not a static hurdle

Our quantitative framework relies on a 9.14% WACC, reflecting an updated capital structure that accounts for $14.3B in total debt and an expanded equity base following the issuance of 36M new Ansys shares in the transaction. The enterprise cost of debt incorporates elevated quarterly interest charges, which stood at $195M in Q4 FY2025 and $133M in Q2 FY2026. The discount rate captures equity market risk while reflecting the operational risk of multi-year software integration.

At a 9.14% hurdle rate, the market's reference spot price of $490.19 prices in sustained cash flow generation that exceeds our modeled terminal parameters. It assumes that cash conversion will accelerate instantaneously without experiencing drag from restructuring initiatives, such as the $118M charge recognized in Q1 FY2026, or annual stock compensation outflows, which totaled $258M in Q1 FY2026 and $232M in Q3 FY2026.

FCF margin normalization is the engine

The transition from a 19.1% free cash flow margin in FY2025 to our modeled terminal targets requires meaningful operational de-bottlenecking. In the base case, the FCF margin expands to 28.0% by FY2031, powered by cost synergies across overlapping administrative functions and operating leverage within Design Automation. The bull case accelerates this expansion to 32.0%, while the bear case limits margin maturation to 24.0% due to persistent integration costs and amortization friction.

The progression of operating margins underscores this divergence. In FY2025, operating income stood at $914.9M on $7.05B of revenue, translating to a 13.0% GAAP operating margin. As non-GAAP adjustments normalize, including quarterly amortization charges exceeding $400M, true cash generation will depend on whether Design IP can sustain its operating margin recovery from the 13.8% low of Q4 FY2025 to levels supporting corporate cash conversion targets.

Mechanics are explicit

Revenue becomes EBIT, cash taxes come out, then we add back D&A, subtract CapEx and the change in net working capital to reach unlevered free cash flow, which we discount and divide by diluted shares.

> **Methodology.** Detailed EBIT-bridge DCF on three scenarios, probability-weighted 25/50/25. Beta decays toward a mature peer over the forecast. Terminal value via Gordon growth. Targets are model-computed, not estimated.

KEY VALUATION PARAMETERS

Forecast window	**FY2026 to FY2031**
Live WACC	**9.14% (2026-10-29)**
Terminal growth	**3.0%**
Diluted shares	**192.3 million**

HOLD

over $341.74 blended target

Competitive moat and concentration risk

We rate the moat as Wide and we view the market structure as Duopoly (EDA). We assign a premium of +30% to sector median multiples on moat strength alone.

- switching costs plus foundry lock-in, designers train for years on a tool flow, and Synopsys co-develops process design kits with TSMC, Samsung, and Intel; the Ansys acquisition (closed July 2025, ~$35B) extends this to multiphysics simulation no rival spans.
- September 30, 2026, Amazon named lead customer for application-optimized silicon IP in a multi-year deal worth more than $1B, shifting to a license-plus-royalty structure (tech-insider, October 2026), a hyperscaler paying royalties is direct pricing-power evidence. Supporting: ~90% time-based/recurring license model and ~$11.0B backlog at April 30, 2026 including $1.8B non-cancellable commitments (Q2 FY2026 10-Q.
- The weakest element is the duopoly rival is a genuine peer (Cadence is stronger in analog/custom), and Ansys integration execution plus export-control exposure to China are live discounts, hence +30%, not the +35% top.

On a multiple cross-check, the justified range is 30.0-33.8x / 22.9-25.8x. The cross-check value is $429-$482 (point $464) vs Base $308.06, spot $466.92, on a base of Framework base from the October 27 moat report. The gap verdict is Market already pays the full wide premium (spot-implied 25.0x at top of range); nothing further available.

Backlog is $10.9bn RPO, 31 Jul 2026, with coverage of 13.5 months vs FY2026 model anchor and visibility rated High. On backlog alone, the stance is Supports, backlog growth trails model growth.

We score concentration and credit risk across seven families, where a High score is 8 points, a Medium score is 4 points, and a Low score is 1 point.

Risk family	Score
customer concentration	1 points
counterparty credit and funding dependence	4 points
single product, single program, or single platform dependency	1 points
single supplier, single fab, or single geography exposure	4 points
platform, channel, or distribution dependency	1 points
customer insourcing risk	4 points
regulatory single point risk	8 points

The combined discount is 20 percent, capped at 20 percent, against an uncapped total of 23 percent. Shown side by side with the moat premium or discount, the net stance is +10 pp points arithmetically. We do not net these figures into a new target, and we do not double count a family excluded under the double-count rule.

The ladder ranks risk coarsely, and because 32 of the 47 names reach the 20 percent cap, the combined discount is not a mechanical valuation haircut until the ladder is recalibrated.

On the gap to spot, the concentration assessment reads as follows. Not at all (corrected from batch arithmetic)

Both the moat premium or discount and the concentration discount are multiple and risk cross-checks dated October 27, 2026, and neither figure is in the DCF target itself. Cross-check figures were computed against the Base in the October 27 framework, so they may differ from a live Base that has since moved.

Business & segment drivers

Revenue prospects anchor on Design Automation Scalability and Ansys Synergies

HOLD

The primary growth engine for Synopsys resides in its Design Automation segment, augmented by the integration of Ansys. In Q1 FY2026, Design Automation revenue surged 96.2% year-over-year to $2.0B, with segment adjusted operating margins reaching 47.3%. This performance continued into Q2 FY2026 with a 62% year-over-year increase, and Q3 FY2026 where the segment generated a 45.2% adjusted operating margin. Management expects Ansys to deliver $2.98B in revenue for FY2026, expanding the addressable domain into multi-physics simulation and system-level verification.

Conversely, the Design IP division has traversed an extended operational reset. After revenue dropped 21% with margins falling to 13.8% in Q4 FY2025, the segment contracted further in Q1 FY2026 with revenue down 6.5% to $407M and margins at 16.2%. Q2 FY2026 experienced a

5.8% decline to $454.2M, with segment operating income declining 26.5%. However, Q3 FY2026 registered an inflection, growing 10.8% year-over-year to $473.8M as segment margins recovered to 26.5%, demonstrating stabilization in high-speed interface and subsystem IP demand.

Top-line visibility is supported by a combined backlog of $11B reported in Q2 FY2026 and $11.4B in Q4 FY2025, providing multi-year license stability. Management raised the FY2026 revenue guidance by $55M at the midpoint to $9.665B during Q2 FY2026, before lifting full-year targets again in Q3 FY2026 across revenue, margins, EPS, and free cash flow. This guidance strength reflects the mission-critical nature of EDA licenses as silicon designers navigate complex architectural transitions.

Nevertheless, divestiture headwinds introduce top-line friction. Management noted a combined ~$150M FY2026 revenue headwind stemming from the divestitures of Optical, PowerArtist, and Processor IP, with the Processor IP divestiture alone accounting for a $40M headwind identified in Q2 FY2026. These portfolio adjustments lower sequential headline momentum while the firm reallocates resources toward core EDA and simulation frameworks.

What to monitor

- **GAAP to Non-GAAP Reconciliation Spread** Track quarterly amortization charges and stock-based compensation, which reached $402M and $232M respectively in Q3 FY2026, preserving a 27-point operating margin divergence.
- **Ansys Revenue Ramp** Monitor execution against the expected $2.98B in FY2026 Ansys revenue as cross-selling synergies develop across Design Automation accounts.
- **Design IP Margin Sustainability** Assess whether segment adjusted operating margins can build upon the 26.5% achieved in Q3 FY2026 after recovering from the 13.8% trough in Q4 FY2025.
- **Deleveraging and Interest Burden** Observe the pace of debt paydown relative to the $14.3B total debt balance and quarterly interest expense levels, which stood at $133M in Q2 FY2026.

+15.1%	77.0%	13.0%	19.1%	$11.4B	4.7%
FY2025 REVENUE GROWTH	GROSS MARGIN	OPERATING MARGIN	FCF MARGIN	NET DEBT	RETURN ON EQUITY

METRIC	FY2025 (AS REPORTED)
Revenue	$7.05B
Gross Profit	$5.43B
EBIT	$914.9M
Net Income	$1.33B
EBITDA	$2.50B
Free Cash Flow	$1.35B
Diluted EPS	$8.04
ROE	4.7%

Risk & moat matrix

The bear case

SHARE EROSION

Organic revenue contraction of ~3% in Q4 FY2025 ex-Ansys demonstrates vulnerability in legacy product lines. Failure to sustain Design IP momentum could allow competitors to capture market share in custom accelerator designs.

M&A DIGESTION

Integrating Ansys introduces substantial overhead, evidenced by $118M in restructuring charges and 10% workforce reductions in Q1 FY2026. Prolonged integration could disrupt core software engineering cycles and delay targeted

REGULATORY AND PRICING SCRUTINY

Divestitures of Optical, PowerArtist, and Processor IP imposed a ~$150M revenue headwind in FY2026. Continued regulatory remedies surrounding large acquisitions could force further portfolio pruning at suboptimal transaction terms.

TECHNOLOGY DISRUPTION

Rapid architectural shifts toward chiplet subsystems initially caused Design IP segment margins to collapse to 13.8% in Q4 FY2025. Further negative operating leverage could emerge if customer design requirements evolve faster than internal IP productization.

REINVESTMENT TREADMILL

Maintaining leadership in chip design automation requires heavy ongoing capitalization, shown by $258M in stock compensation in Q1 FY2026. Escalating software development costs could impede the projected expansion toward a 28.0% base-case FCF margin.

WACC AND DURATION SENSITIVITY

With an enterprise net debt load of $11.4B and a high discount rate of 9.14%, equity valuation is acutely sensitive to cost of capital fluctuations. Prolonged high borrowing costs directly reduce equity value via persistent interest charges.

operational synergies.

The strengths

OLIGOPOLY POSITION

Synopsys remains the dominant chip design automation software vendor selling licenses chipmakers need to design silicon. Extremely high switching costs ensure that flagship semiconductor projects cannot easily substitute core verification tools.

FCF COMPOUNDING

Operating cash flow rebounded to $856.8M in Q1 FY2026, reversing prior-year outflows of $67.5M. This underlying cash generation supports a $2.0B board-authorized repurchase program while systematically reducing balance sheet debt.

GEOGRAPHIC DURABILITY

Global semiconductor fabrication across complex nodes relies universally on Synopsys design licenses and subsystem IP blocks. This broad geographic necessity provides stability against regional capex cyclicality in the broader hardware sector.

PIPELINE OPTIONALITY

The multi-year backlog stood at $11B in Q2 FY2026 and $11.4B in Q4 FY2025, offering predictable long-term cash flow visibility. The addition of Ansys positions the company to capture expanding multi-physics simulation budgets across advanced system designs.

The bottom line on risk and reward

Synopsys presents an exceptional software moat married to an extended valuation profile. The strategic rationale behind combining Design Automation with Ansys multi-physics simulation tools is sound, as evidenced by segment adjusted operating margins of 45.2% and the expectation of $2.98B in Ansys revenue for FY2026. The Q3 FY2026 beat-and-raise quarter confirmed operational resilience, with revenue reaching $2.48B, non-GAAP EPS hitting $3.91, and Design IP returning to 10.8% growth.

However, fundamental reality diverges sharply from current equity pricing. The enterprise carries $14.3B in total debt and $11.4B in net debt, yielding significant quarterly interest expense that stood at $133M in Q2 FY2026 and $195M in Q4 FY2025. Furthermore, the true GAAP economic profitability of the firm remains obscured by heavy ongoing charges, including $402M in amortization and $232M in stock-based compensation in Q3 FY2026 alone. At an EV to latest fiscal year EBITDA multiple of 42.3x and a P/E multiple of 61.0x on FY2025 diluted EPS of $8.04, the market has priced in seamless multi-year synergy execution without factoring in potential macro or integration hazards.

Our probability-weighted framework derives an intrinsic value of $341.74 across 192.3 million diluted shares, indicating -30.3% upside relative to the reference spot price of $490.19. We maintain our Hold (Valuation Concerns) rating, advising institutional investors to await a more

attractive entry point that adequately reflects post-acquisition integration risks, balance sheet leverage, and capital structure dilution.

Justin Wen, Senior Equity Research Analyst, Cascadia Analytics

Franchise deep dive

Design Automation and EDA Software Licenses: Mission-Critical Silicon Infrastructure

The foundation of Synopsys is its dominant chip design automation software suite, which provides the licenses chipmakers need to design silicon. The tools automate the complex journey from algorithmic concept to physical silicon masks. Without these automated toolchains, modern semiconductor fabrication across nanoscale process geometries would be mathematically impossible. The software handles logic synthesis, place-and-route, physical verification, and timing closure, embedding Synopsys directly into customer research and development budgets.

The strategic addition of Ansys elevates this franchise into multi-physics simulation, encompassing thermal, mechanical, and electromagnetic verification. In Q1 FY2026, Design Automation segment revenue surged 96.2% year-over-year to $2.0B with adjusted operating margins expanding to 47.3%. In Q2 FY2026, segment revenue expanded 62% year-over-year, and in Q3 FY2026, the unit sustained an adjusted operating margin of 45.2%. This performance confirms that system-level complexity continues to drive robust adoption across the EDA portfolio.

Simultaneously, the Design IP division provides validated building blocks that integrate into the core software workflows. Although the segment experienced notable headwinds, with revenue declining 21% in Q4 FY2025 and operating margin dropping to 13.8%, Q3 FY2026 confirmed an operational inflection. Design IP revenue rebounded 10.8% year-over-year to $473.8M, supported by segment operating margins recovering to 26.5%. The synergy between proprietary EDA licenses and plug-and-play IP blocks creates a cohesive ecosystem that locks in customer development teams across successive architectural nodes.

What the recent record shows

- Design Automation revenue expanded 96.2% year-over-year to $2.0B in Q1 FY2026 and 62% in Q2 FY2026, propelled by Ansys integration and AI silicon demand.
- Design Automation adjusted operating margins sustained elite levels, recording 41.5% in Q4 FY2025, 47.3% in Q1 FY2026, and 45.2% in Q3 FY2026.
- Design IP experienced an operational turnaround in Q3 FY2026, rising 10.8% year-over-year to $473.8M with adjusted operating margins recovering to 26.5% from 13.8% in Q4 FY2025.

- Combined enterprise backlog remained substantial at $11.4B in Q4 FY2025 and $11B in Q2 FY2026, underwriting multi-year license visibility.

Our read: Synopsys retains an unrivaled competitive moat in core EDA and Design Automation, but rich trading multiples of 61.0x P/E and 42.3x EV to EBITDA fully capture this franchise strength. With $14.3B in total debt and ongoing GAAP-to-non-GAAP differentials, we recommend maintaining a Hold position until valuation parameters align with our $341.74 blended target.